Titanic EDA

Exploratory Data Analysis - Titanic Passenger Survival Dataset

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Subject: Data Science

Exploratory Data Analysis (EDA)

Titanic Passenger Survival Dataset

Course: Introduction to Data Science

Assignment: Complete EDA following the Lecture 6 flow

Section	Content
1	Dataset Introduction
2	Univariate EDA
3	Target Analysis
4	Bivariate EDA
5	Correlation Analysis
6	Multivariate EDA
7	Patterns, Trends and Anomalies
8	Final Findings

Tools: Python, pandas, NumPy, Matplotlib, Seaborn (Jupyter Notebook). The code, outputs and graphs below are taken directly from the executed notebook.

Assignment Statement

Select a real-world dataset from Kaggle, UCI or another public source and perform a complete Exploratory Data Analysis (EDA) to understand its distributions, relationships, patterns and unusual observations, following the Lecture 6 flow: understand the dataset and target, explore individual features, compare variables, study relationships, examine multiple features together, and summarize meaningful insights.

Tasks

1. Dataset Introduction: title and source, rows and columns, main features, target variable, problem/question.
2. Univariate EDA: histogram, density plot, bar chart, boxplot with brief interpretation.
3. Target Analysis: distribution, balanced or imbalanced, brief explanation.
4. Bivariate EDA: scatter plot, feature vs target, grouped/stacked bar chart, boxplot by category.
5. Correlation Analysis: heatmap, strong positive/negative and weak relationships, interpretation.
6. Multivariate EDA: combined patterns, groups, relationships.
7. Patterns, Trends and Anomalies.
8. Final Findings: 5-8 key insights and how they help future modelling.

Submission Requirements

- Notebook containing code, outputs, visualizations, short interpretation after important graphs, and final findings.
- Upload the completed work to GitHub or Zenodo and submit the public link only on LMS.

Where each task is covered in this document

Task	Section in this report
1	Section 1: Dataset Introduction
2	Section 2: Univariate EDA (2.1 numerical, 2.2 categorical)
3	Section 3: Target Analysis
4	Section 4: Bivariate EDA (4.1 to 4.3)
5	Section 5: Correlation Analysis
6	Section 6: Multivariate EDA (6.1 to 6.3)
7	Section 7: Patterns, Trends and Anomalies
8	Section 8: Final Findings

Code

```
import warnings; warnings.filterwarnings("ignore")
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid", palette="deep")
plt.rcParams.update({"figure.dpi": 100, "axes.titlesize": 12, "axes.labelsize": 10})

URL = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
df = pd.read_csv(URL)          # if offline: pd.read_csv("titanic.csv")
df.head()
```

Output

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

1. Dataset Introduction

Item	Details
Title	Titanic - Passenger Survival Dataset
Source	Kaggle "Titanic - Machine Learning from Disaster" (https://www.kaggle.com/c/titanic). The copy used here is the cleaned version distributed with the Seaborn library (https://github.com/mwaskom/seaborn-data)
Rows x Columns	891 passengers x 15 columns (verified in the code below)
Target variable	`survived` (0 = did not survive, 1 = survived)
Problem / question	*Which passenger characteristics (sex, ticket class, age, fare, family size, port of embarkation) are associated with surviving the Titanic disaster?*

Main features

Feature	Type	Meaning
survived	Binary (target)	1 = survived, 0 = died
pclass / class	Ordinal	Ticket class (1 = First, 2 = Second, 3 = Third)
sex	Categorical	male / female
age	Numerical	Age in years (has missing values)
sibsp	Numerical (count)	Number of siblings / spouses aboard
parch	Numerical (count)	Number of parents / children aboard
fare	Numerical	Ticket price (pounds)
embarked / embark_town	Categorical	Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)
who, adult_male	Categorical	man / woman / child (derived from age and sex)
deck	Categorical	Cabin deck letter (mostly missing)
alive	Categorical	Text version of the target (yes / no)
alone	Boolean	True if the passenger travelled with no family

Code

```
print("Shape (rows, columns):", df.shape)
print("\nColumn data types:")
print(df.dtypes)
print("\nDuplicate rows:", df.duplicated().sum())
```

Output

```
Shape (rows, columns): (891, 15)

Column data types:
survived         int64
pclass           int64
sex                str
age            float64
sibsp            int64
parch            int64
fare           float64
embarked           str
class              str
who                str
adult_male        bool
deck               str
embark_town        str
alive              str
alone             bool
dtype: object

Duplicate rows: 107
```

Code

```
missing = pd.DataFrame({"missing_count": df.isna().sum(),
                        "missing_%": (df.isna().mean()*100).round(1)})
missing[missing.missing_count > 0].sort_values("missing_count", ascending=False)
```

Output

```
             missing_count  missing_%
deck                   688       77.2
age                    177       19.9
embarked                 2        0.2
embark_town              2        0.2
```

Code

```
df.describe().round(2)
```

Output

```
       survived  pclass     age   sibsp   parch    fare
count    891.00  891.00  714.00  891.00  891.00  891.00
mean       0.38    2.31   29.70    0.52    0.38   32.20
std        0.49    0.84   14.53    1.10    0.81   49.69
min        0.00    1.00    0.42    0.00    0.00    0.00
25%        0.00    2.00   20.12    0.00    0.00    7.91
50%        0.00    3.00   28.00    0.00    0.00   14.45
75%        1.00    3.00   38.00    1.00    0.00   31.00
max        1.00    3.00   80.00    8.00    6.00  512.33
```

Data quality observations.

- `deck` is missing for 688 of 891 passengers (77.2%) and `age` for 177 (19.9%); `embarked` / `embark_town` miss only 2 values.
- The 107 duplicate rows are *not* data-entry errors: the table has no passenger ID column, so different passengers with identical attributes (for example two third-class men aged 22 paying the same fare) look identical. All 891 rows are therefore kept.
- `class`, `alive` and `embarked` repeat information from `pclass`, `survived` and `embark_town`, so only one of each pair is needed in analysis.
- Descriptive statistics already hint at skewness: mean fare (32.2) is more than double the median fare (14.45), and maximum fare is 512.33.

Code

```
# Light feature preparation used later in the EDA (original df is kept untouched)
d = df.copy()
d["family_size"] = d["sibsp"] + d["parch"] + 1
d["age_group"] = pd.cut(d["age"], bins=[0, 12, 18, 35, 60, 100],
                        labels=["Child (0-12)", "Teen (13-18)", "Adult (19-35)", "Middle (36-60)",
"Senior (60+)"])
d["has_cabin_info"] = d["deck"].notna()
d["age_missing"] = d["age"].isna()
d["log_fare"] = np.log1p(d["fare"])
d["sex_pclass"] = d["sex"] + " - class " + d["pclass"].astype(str)
d[["family_size", "age_group", "has_cabin_info", "log_fare"]].head()
```

Output

```
   family_size       age_group  has_cabin_info  log_fare
```

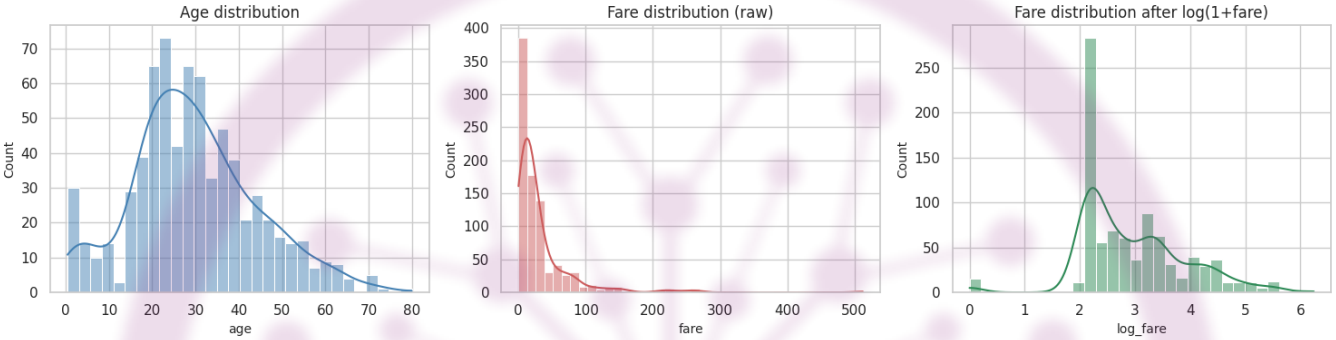
```
0            2   Adult (19-35)          False  2.110213
1            2  Middle (36-60)           True  4.280593
2            1   Adult (19-35)          False  2.188856
3            2   Adult (19-35)           True  3.990834
4            1   Adult (19-35)          False  2.202765
```

2. Univariate EDA

2.1 Numerical features: histogram + density, and boxplot

Code

```
fig, ax = plt.subplots(1, 3, figsize=(15, 4))
sns.histplot(d["age"].dropna(), bins=30, kde=True, ax=ax[0], color="steelblue")
ax[0].set_title("Age distribution")
sns.histplot(d["fare"], bins=40, kde=True, ax=ax[1], color="indianred")
ax[1].set_title("Fare distribution (raw)")
sns.histplot(d["log_fare"], bins=30, kde=True, ax=ax[2], color="seagreen")
ax[2].set_title("Fare distribution after log(1+fare)")
plt.tight_layout(); plt.show()
print("Skewness  -> age: %.2f | fare: %.2f | log_fare: %.2f" % (d.age.skew(), d.fare.skew(),
d.log_fare.skew()))
```



Output

```
Skewness  -> age: 0.39 | fare: 4.79 | log_fare: 0.39
```

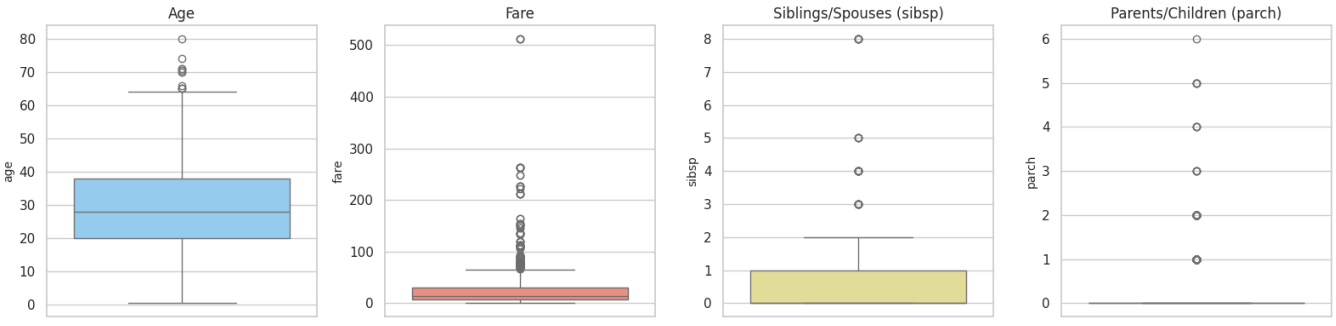
Interpretation.

- **Age** is roughly bell-shaped (skewness 0.39), centred near 28 years with a small extra peak of infants and toddlers under 5. Most passengers are young adults, and few are above 60.
- **Fare** is extremely right-skewed (skewness 4.79): most tickets cost under 31 pounds but a long tail runs up to 512. This is why the mean is far above the median.
- After the `log(1+fare)` transformation the skewness drops to 0.39, so the log version is much better suited for linear/distance-based models.

Code

```
fig, ax = plt.subplots(1, 4, figsize=(16, 4))
sns.boxplot(y=d["age"], ax=ax[0], color="lightskyblue");  ax[0].set_title("Age")
sns.boxplot(y=d["fare"], ax=ax[1], color="salmon");       ax[1].set_title("Fare")
sns.boxplot(y=d["sibsp"], ax=ax[2], color="khaki");       ax[2].set_title("Siblings/Spouses (sibsp)")
sns.boxplot(y=d["parch"], ax=ax[3], color="plum");        ax[3].set_title("Parents/Children (parch)")
plt.tight_layout(); plt.show()

for c in ["age", "fare"]:
    q1, q2, q3 = d[c].quantile([.25, .5, .75]); iqr = q3 - q1
    lo, hi = q1 - 1.5*iqr, q3 + 1.5*iqr
    n_out = ((d[c] < lo) | (d[c] > hi)).sum()
    print(f"{c:5s}: Q1={q1:.2f}  median={q2:.2f}  Q3={q3:.2f}  IQR={iqr:.2f}  upper fence={hi:.2f}
outliers={n_out}")
```



Output

```
age  : Q1=20.12  median=28.00  Q3=38.00  IQR=17.88  upper fence=64.81  outliers=11
fare : Q1=7.91  median=14.45  Q3=31.00  IQR=23.09  upper fence=65.63  outliers=116
```

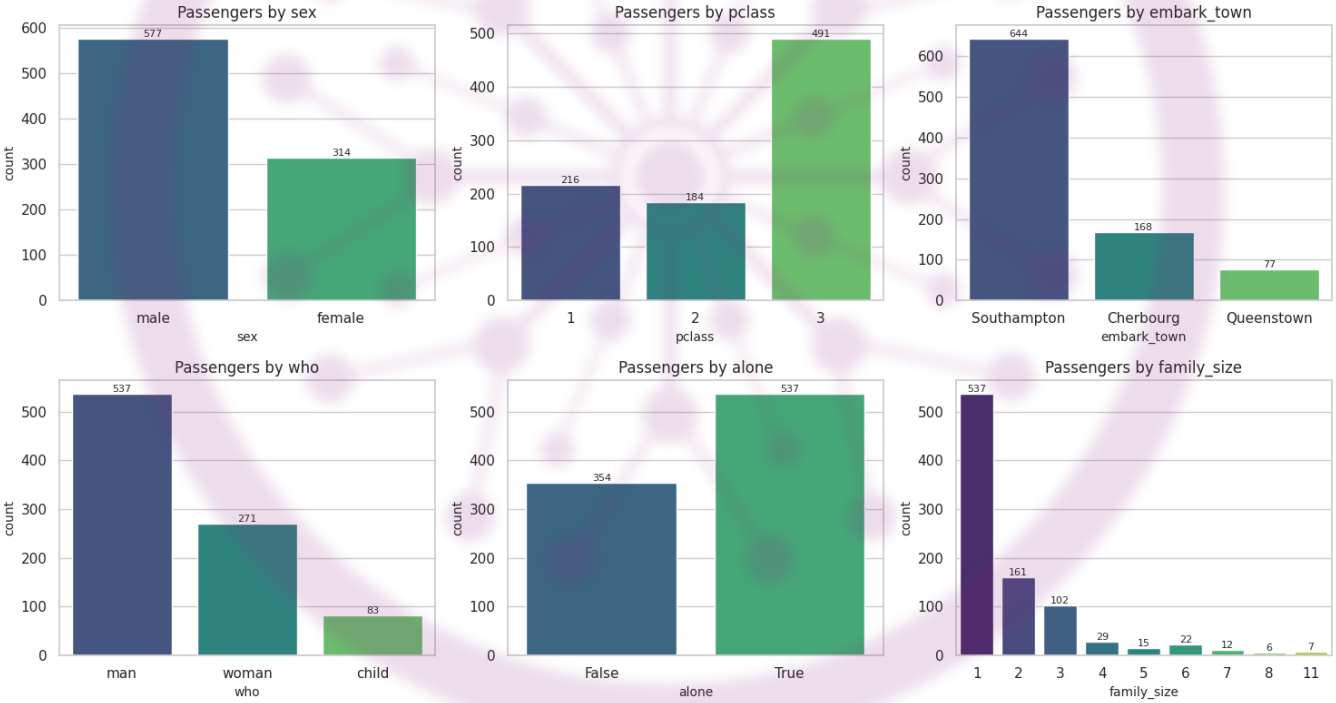
Interpretation.

- **Age:** median 28, middle 50% of passengers between 20 and 38. Only 11 passengers are flagged as outliers (older than about 65), which is a natural, valid group of elderly travellers.
- **Fare:** median 14.45 with Q3 of 31; the upper fence is about 65.6 and **116 passengers** lie above it. Many of these are genuine first-class fares, so they should be treated carefully, not deleted.
- **sibsp / parch:** the box collapses to 0 because most people travelled without relatives. The dots above are rare large families (up to 8 siblings/spouses).

2.2 Categorical features: bar charts

Code

```
fig, ax = plt.subplots(2, 3, figsize=(15, 8))
order_cfg = [("sex", ax[0,0]), ("pclass", ax[0,1]), ("embark_town", ax[0,2]),
             ("who", ax[1,0]), ("alone", ax[1,1]), ("family_size", ax[1,2])]
for col, a in order_cfg:
    sns.countplot(data=d, x=col, ax=a, palette="viridis")
    a.set_title(f"Passengers by {col}")
    for p in a.patches:
        a.annotate(int(p.get_height()), (p.get_x()+p.get_width()/2, p.get_height()),
                   ha="center", va="bottom", fontsize=8)
plt.tight_layout(); plt.show()
print(d["sex"].value_counts(normalize=True).round(3).to_dict())
print(d["pclass"].value_counts(normalize=True).round(3).to_dict())
print(d["embark_town"].value_counts(normalize=True).round(3).to_dict())
```



Output

```
{'male': 0.648, 'female': 0.352}
{3: 0.551, 1: 0.242, 2: 0.207}
{'Southampton': 0.724, 'Cherbourg': 0.189, 'Queenstown': 0.087}
```

Interpretation.

- **Sex:** 577 men (64.8%) vs 314 women (35.2%): men outnumber women almost 2 to 1.
- **Class:** Third class is the largest group (491, 55.1%), followed by First (216, 24.2%) and Second (184, 20.7%).
- **Port:** Southampton dominates with 644 passengers (72.4%); Cherbourg has 168 and Queenstown only 77, so Queenstown estimates will be less reliable.
- **Who / alone:** 537 passengers are adult men, 271 women and 83 children. 537 passengers (60.3%) travelled alone.

- **Family size:** size 1 (alone) is by far the most common; groups larger than 4 are very rare (few observations, so rates for them are noisy).

3. Target Analysis (`survived`)

Code

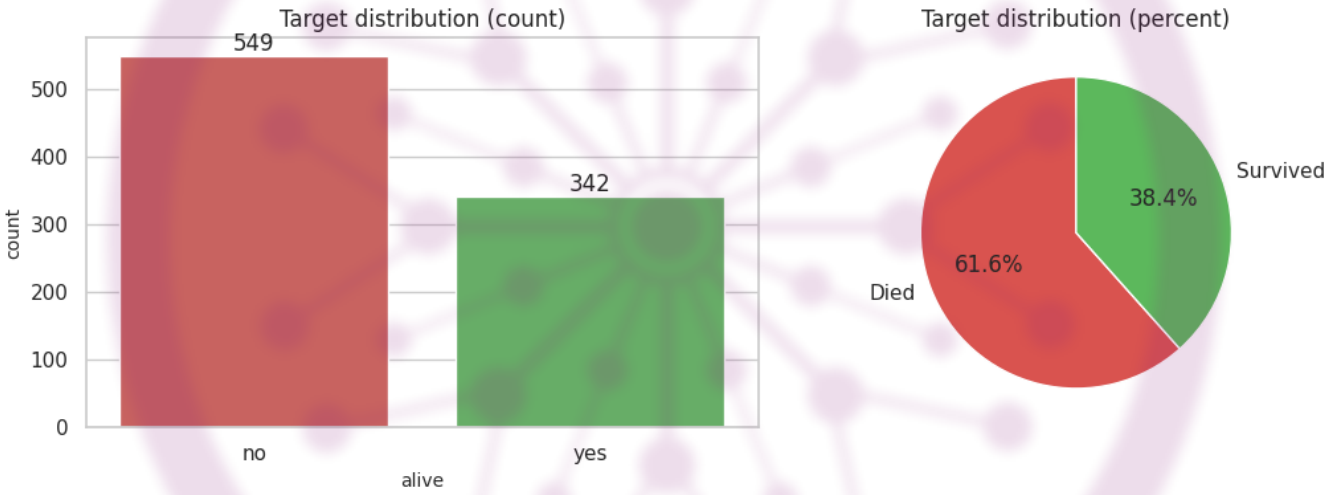
```
counts = d["survived"].value_counts().sort_index()
perc = (d["survived"].value_counts(normalize=True).sort_index()*100).round(2)
print(pd.DataFrame({"count": counts, "percent": perc}).rename(index={0: "0 (died)", 1: "1 (survived)"}))
print("\nImbalance ratio (majority : minority) = %.2f : 1" % (counts.max()/counts.min()))

fig, ax = plt.subplots(1, 2, figsize=(11, 4))
sns.countplot(data=d, x="alive", order=["no", "yes"], ax=ax[0], palette=["#d9534f", "#5cb85c"])
for p in ax[0].patches:
    ax[0].annotate(int(p.get_height()), (p.get_x()+p.get_width()/2, p.get_height()), ha="center",
va="bottom")
ax[0].set_title("Target distribution (count)")
ax[1].pie(counts.values, labels=["Died", "Survived"], autopct="%1.1f%%", colors=["#d9534f", "#5cb85c"],
startangle=90)
ax[1].set_title("Target distribution (percent)")
plt.tight_layout(); plt.show()
```

Output

```
              count  percent
survived
0 (died)        549    61.62
1 (survived)    342    38.38

Imbalance ratio (majority : minority) = 1.61 : 1
```



Observation. 549 passengers (61.6%) died and 342 (38.4%) survived, an imbalance ratio of about **1.6 : 1**. This is a **mild imbalance**: not balanced perfectly, but not severe (like 95/5 fraud data). Accuracy alone would still look decent (a model that always predicts "died" gets 61.6%), so in modelling we should also report precision, recall, F1 and ROC-AUC, and use stratified train/test splits. Class weights can be tried but heavy resampling is not needed.

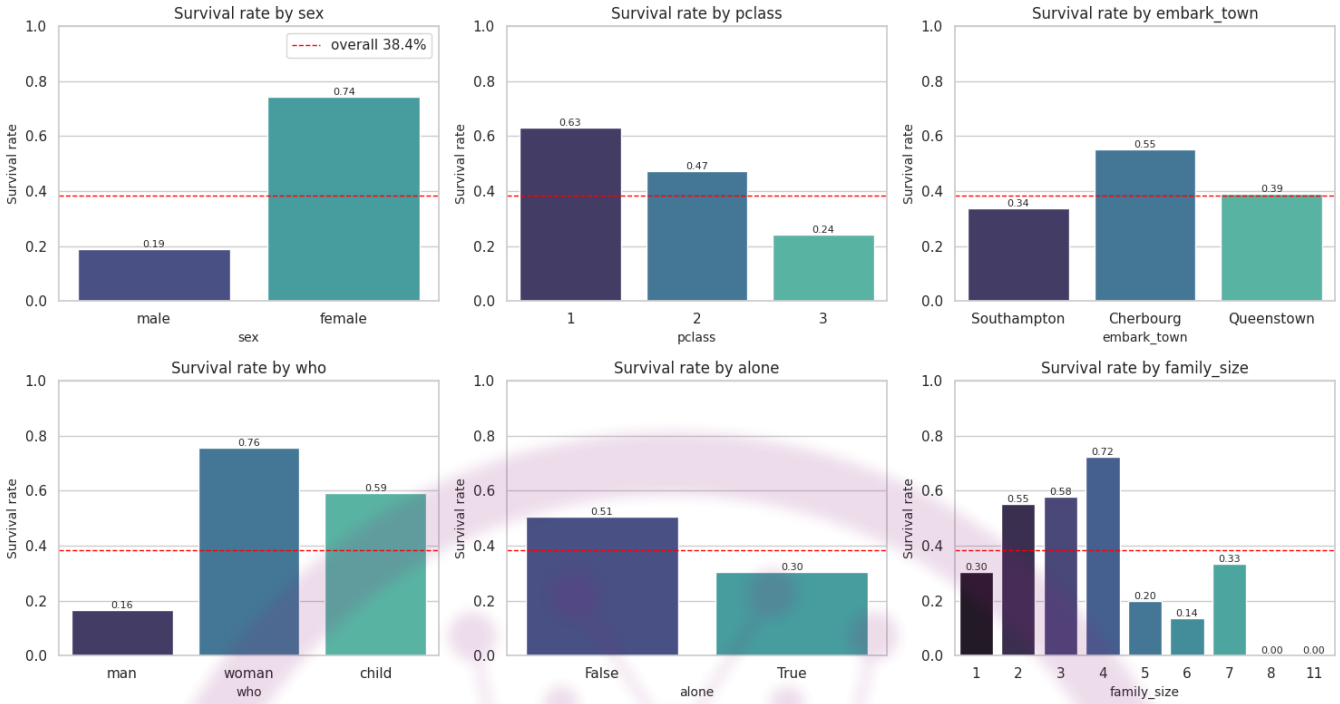
4. Bivariate EDA

4.1 Feature vs target: survival rate by category

Code

```
fig, ax = plt.subplots(2, 3, figsize=(15, 8))
for col, a in [("sex", ax[0,0]), ("pclass", ax[0,1]), ("embark_town", ax[0,2]),
               ("who", ax[1,0]), ("alone", ax[1,1]), ("family_size", ax[1,2])]:
    sns.barplot(data=d, x=col, y="survived", ax=a, palette="mako", errorbar=None)
    a.axhline(d.survived.mean(), color="red", ls="--", lw=1, label="overall 38.4%")
    a.set_ylim(0, 1); a.set_ylabel("Survival rate"); a.set_title(f"Survival rate by {col}")
    for p in a.patches:
        a.annotate(f"{p.get_height():.2f}", (p.get_x()+p.get_width()/2, p.get_height()), ha="center",
va="bottom", fontsize=8)
ax[0,0].legend()
plt.tight_layout(); plt.show()

for c in ["sex", "pclass", "embark_town", "who", "alone"]:
    print(d.groupby(c)["survived"].agg(["mean", "count"]).round(3), "\n")
```



Output

```
          mean  count
sex
female   0.742    314
male     0.189    577

          mean  count
pclass
1        0.630    216
2        0.473    184
3        0.242    491

              mean  count
embark_town
Cherbourg    0.554    168
Queenstown   0.390     77
Southampton  0.337    644

         mean  count
who
child   0.590     83
man     0.164    537
woman   0.756    271

         mean  count
alone
False   0.506    354
True    0.304    537
```

Interpretation.

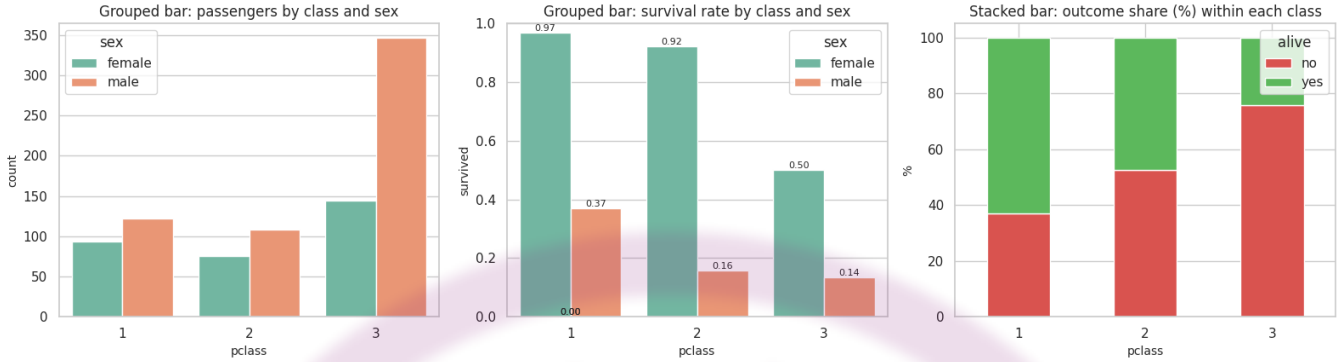
- **Sex is the strongest single separator:** 74.2% of women survived versus only 18.9% of men.
- **Class matters strongly and monotonically:** survival is 63.0% (First), 47.3% (Second) and 24.2% (Third).
- **Port:** Cherbourg passengers survived most (55.4%) vs Queenstown (39.0%) and Southampton (33.7%). This is probably indirect: Cherbourg had a higher share of first-class passengers.
- **Who:** women 75.6%, children 59.0%, men 16.4% - matches the "women and children first" evacuation pattern.
- **Alone:** passengers travelling with family survived more (50.6%) than those alone (30.4%), but very large families (5+ members) did poorly, so the relationship with family size is non-linear.

4.2 Grouped bar chart and stacked bar chart

Code

```
fig, ax = plt.subplots(1, 3, figsize=(16, 4.5))
sns.countplot(data=d, x="pclass", hue="sex", ax=ax[0], palette="Set2")
ax[0].set_title("Grouped bar: passengers by class and sex")
sns.barplot(data=d, x="pclass", y="survived", hue="sex", ax=ax[1], palette="Set2", errorbar=None)
ax[1].set_title("Grouped bar: survival rate by class and sex"); ax[1].set_ylim(0, 1)
```

```
for p in ax[1].patches:
    ax[1].annotate(f"{p.get_height():.2f}", (p.get_x()+p.get_width()/2, p.get_height()), ha="center",
va="bottom", fontsize=8)
ct = pd.crosstab(d["pclass"], d["alive"], normalize="index")[["no", "yes"]]*100
ct.plot(kind="bar", stacked=True, ax=ax[2], color=["#d9534f", "#5cb85c"], rot=0)
ax[2].set_title("Stacked bar: outcome share (%) within each class"); ax[2].set_ylabel("%")
plt.tight_layout(); plt.show()
print(ct.round(1))
```



Output

```
alive     no   yes
pclass
1       37.0  63.0
2       52.7  47.3
3       75.8  24.2
```

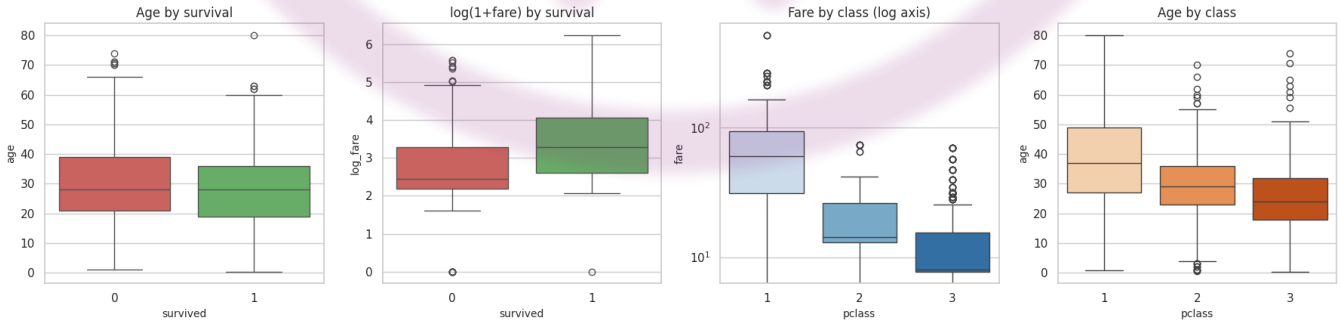
Interpretation.

- Third class is dominated by men (347 men vs 144 women), which partly explains its low overall survival.
- Within each class, women survive far more than men: First 96.8% vs 36.9%, Second 92.1% vs 15.7%, Third 50.0% vs 13.5%. So **sex and class both matter, and sex matters inside every class**.
- Third-class women (50%) did much worse than first/second-class women (above 90%), showing a strong class effect even among women.

4.3 Boxplots by category / target, and scatter plot

Code

```
fig, ax = plt.subplots(1, 4, figsize=(18, 4.5))
sns.boxplot(data=d, x="survived", y="age", ax=ax[0], palette=["#d9534f", "#5cb85c"]);
ax[0].set_title("Age by survival")
sns.boxplot(data=d, x="survived", y="log_fare", ax=ax[1], palette=["#d9534f", "#5cb85c"]);
ax[1].set_title("log(1+fare) by survival")
sns.boxplot(data=d, x="pclass", y="fare", ax=ax[2], palette="Blues"); ax[2].set_yscale("log");
ax[2].set_title("Fare by class (log axis)")
sns.boxplot(data=d, x="pclass", y="age", ax=ax[3], palette="Oranges"); ax[3].set_title("Age by class")
plt.tight_layout(); plt.show()
print(d.groupby("survived")[["age", "fare"]].median().round(2))
print(d.groupby("pclass")[["age", "fare"]].median().round(2))
```



Output

```
          age  fare
survived
0        28.0  10.5
1        28.0  26.0
         age   fare
pclass
1       37.0  60.29
2       29.0  14.25
```

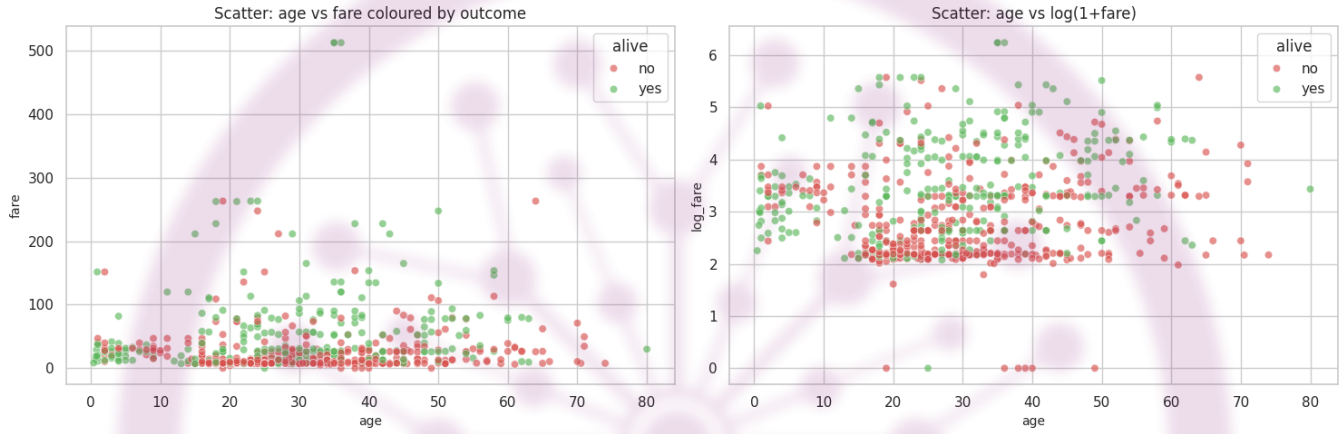
```
3       24.0   8.05
```

Interpretation.

- Survivors and non-survivors have very similar age distributions (median differs only slightly), so age alone is a weak predictor. Its effect appears mainly among children.
- Survivors paid clearly higher fares (log-fare box is shifted up) - fare is acting as a proxy for class.
- Fare medians by class are 60.29 (First), 14.25 (Second) and 8.05 (Third), with wide spread in First class. Median age also falls with class: 37 (First), 29 (Second), 24 (Third).

Code

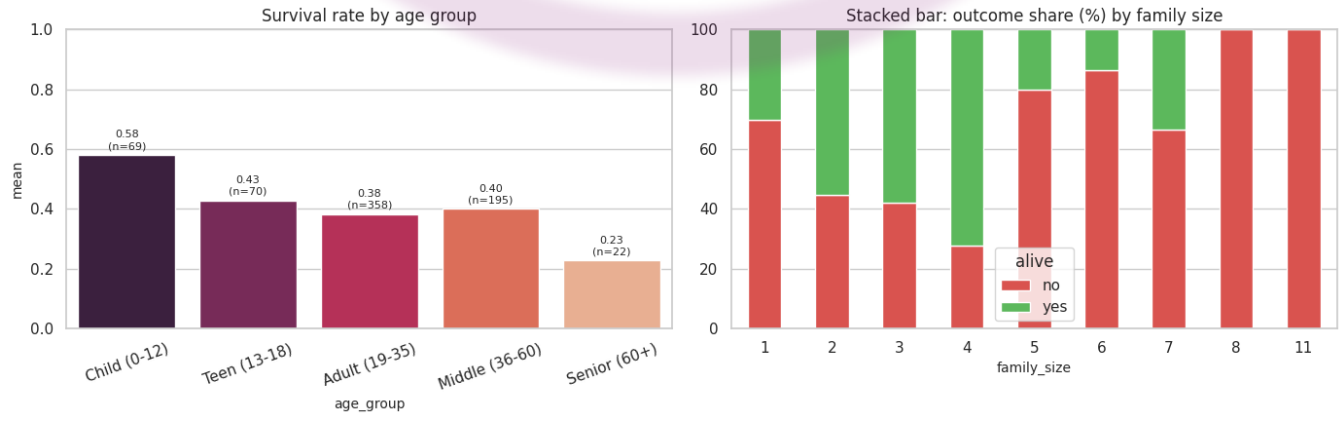
```
fig, ax = plt.subplots(1, 2, figsize=(15, 5))
sns.scatterplot(data=d, x="age", y="fare", hue="alive", hue_order=["no", "yes"],
                palette=["#d9534f", "#5cb85c"], alpha=0.65, ax=ax[0])
ax[0].set_title("Scatter: age vs fare coloured by outcome")
sns.scatterplot(data=d, x="age", y="log_fare", hue="alive", hue_order=["no", "yes"],
                palette=["#d9534f", "#5cb85c"], alpha=0.65, ax=ax[1])
ax[1].set_title("Scatter: age vs log(1+fare)")
plt.tight_layout(); plt.show()
```



Interpretation. There is no clear linear relationship between age and fare (correlation about 0.10). Survivors (green) are more frequent at high fares, and a visible cluster of very young survivors appears at the left edge. The log scale spreads out the crowded low-fare region and makes this clearer. The few points at 512 are extreme but belong to the same first-class group.

Code

```
fig, ax = plt.subplots(1, 2, figsize=(14, 4.5))
age_surv = d.groupby("age_group")["survived"].agg(["mean", "count"])
sns.barplot(x=age_surv.index, y=age_surv["mean"], ax=ax[0], palette="rocket")
ax[0].set_title("Survival rate by age group"); ax[0].set_ylim(0, 1); ax[0].tick_params(axis="x",
rotation=20)
for i, (m, n) in enumerate(zip(age_surv["mean"], age_surv["count"])):
    ax[0].text(i, m + 0.02, f"{m:.2f}\n(n={n})", ha="center", fontsize=8)
fs = pd.crosstab(d["family_size"], d["alive"], normalize="index")[["no", "yes"]]*100
fs.plot(kind="bar", stacked=True, ax=ax[1], color=["#d9534f", "#5cb85c"], rot=0)
ax[1].set_title("Stacked bar: outcome share (%) by family size")
plt.tight_layout(); plt.show()
print(age_surv.round(3))
```



Output

```
                 mean  count
age_group
Child (0-12)    0.580     69
Teen (13-18)    0.429     70
Adult (19-35)   0.383    358
Middle (36-60)  0.400    195
Senior (60+)    0.227     22
```

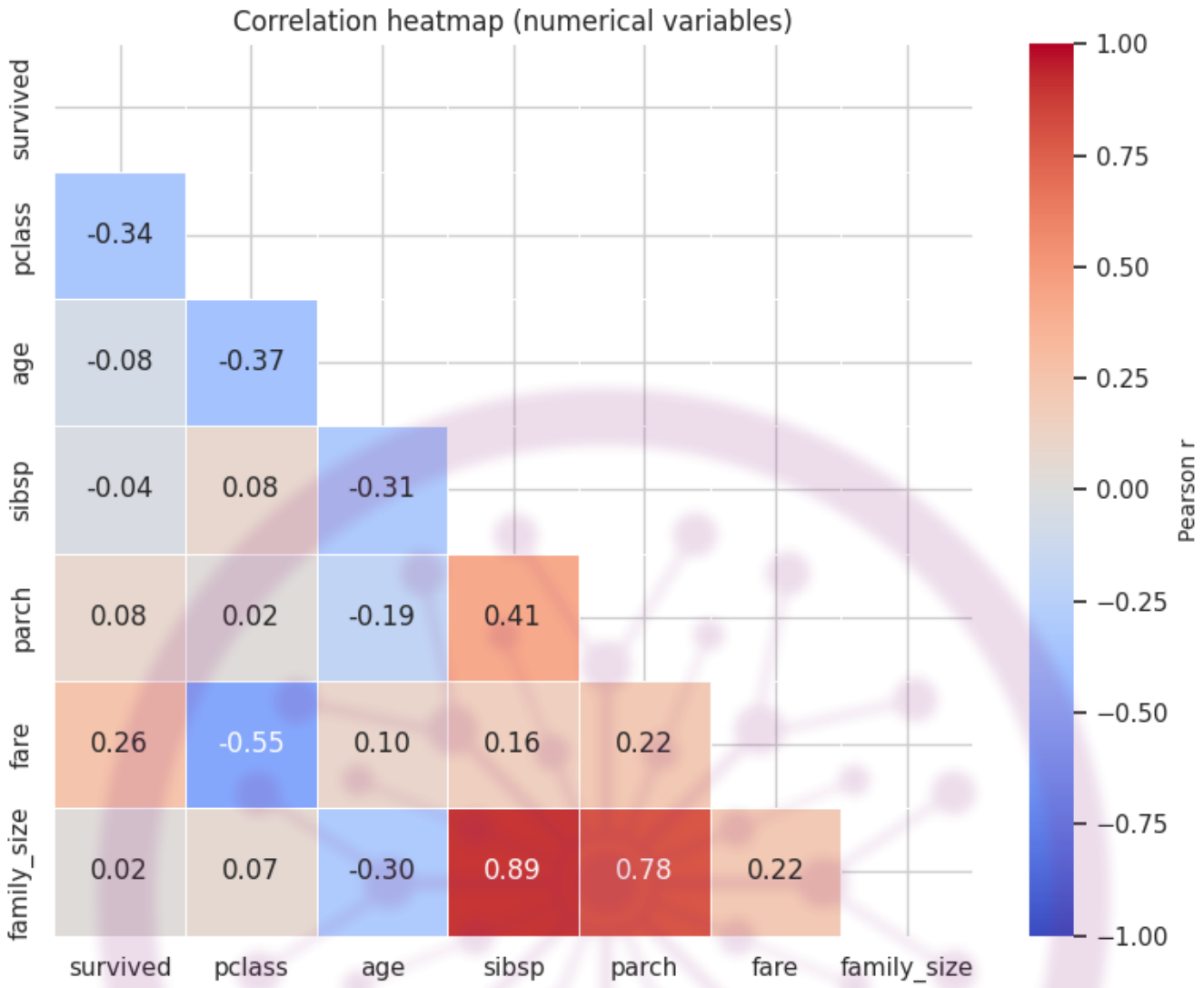
Interpretation. Children aged 0-12 survived at 58.0%, the highest of all age groups, while passengers above 60 survived at only 22.7% (n = 22). Teens, adults and middle-aged passengers sit near 38-43%. For family size, passengers in groups of 2-4 survived most (55-72%), while those alone (30.4%) and in groups of 5+ (0-33%) did worse - a **non-linear** pattern that a plain linear term would miss.

5. Correlation Analysis

Code

```
num_cols = ["survived", "pclass", "age", "sibsp", "parch", "fare", "family_size"]
corr = d[num_cols].corr()
plt.figure(figsize=(8, 6.5))
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0, vmin=-1, vmax=1,
            mask=mask, linewidths=.5, cbar_kws={"label": "Pearson r"})
plt.title("Correlation heatmap (numerical variables)")
plt.tight_layout(); plt.show()

pairs = (corr.where(~mask).stack().rename("r").reset_index()
             .query("level_0 != level_1"))
pairs.columns = ["var1", "var2", "r"]
pairs["abs_r"] = pairs["r"].abs()
print("Top pairs by absolute correlation:")
print(pairs.sort_values("abs_r",
ascending=False).head(8).drop(columns="abs_r").round(2).to_string(index=False))
print("\nCorrelation with target (survived):")
print(corr["survived"].drop("survived").sort_values().round(2))
```



Output

```
Top pairs by absolute correlation:
       var1     var2     r
family_size    sibsp  0.89
family_size    parch  0.78
       fare   pclass -0.55
      parch    sibsp  0.41
        age   pclass -0.37
     pclass survived -0.34
      sibsp      age -0.31
family_size      age -0.30

Correlation with target (survived):
pclass        -0.34
age           -0.08
sibsp         -0.04
family_size    0.02
parch          0.08
fare           0.26
Name: survived, dtype: float64
```

Interpretation.

- **Strongest negative:** `pclass` vs `fare` (r = -0.55): higher class number (cheaper class) means lower fare. `pclass` vs `survived` (-0.34) and `pclass` vs `age` (-0.37) are moderate negatives.
- **Strong/moderate positive:** `fare` vs `survived` (+0.26) is a modest positive; `sibsp` vs `parch` (+0.41) is the largest positive pair among the original features, because families travel together (`family_size` is built from both, so its r of 0.89 and 0.78 with them is by construction).
- **Weak:** `age` vs `survived` (-0.08), `sibsp` vs `survived` (-0.04), `parch` vs `survived` (+0.08), `age` vs `fare` (+0.10).

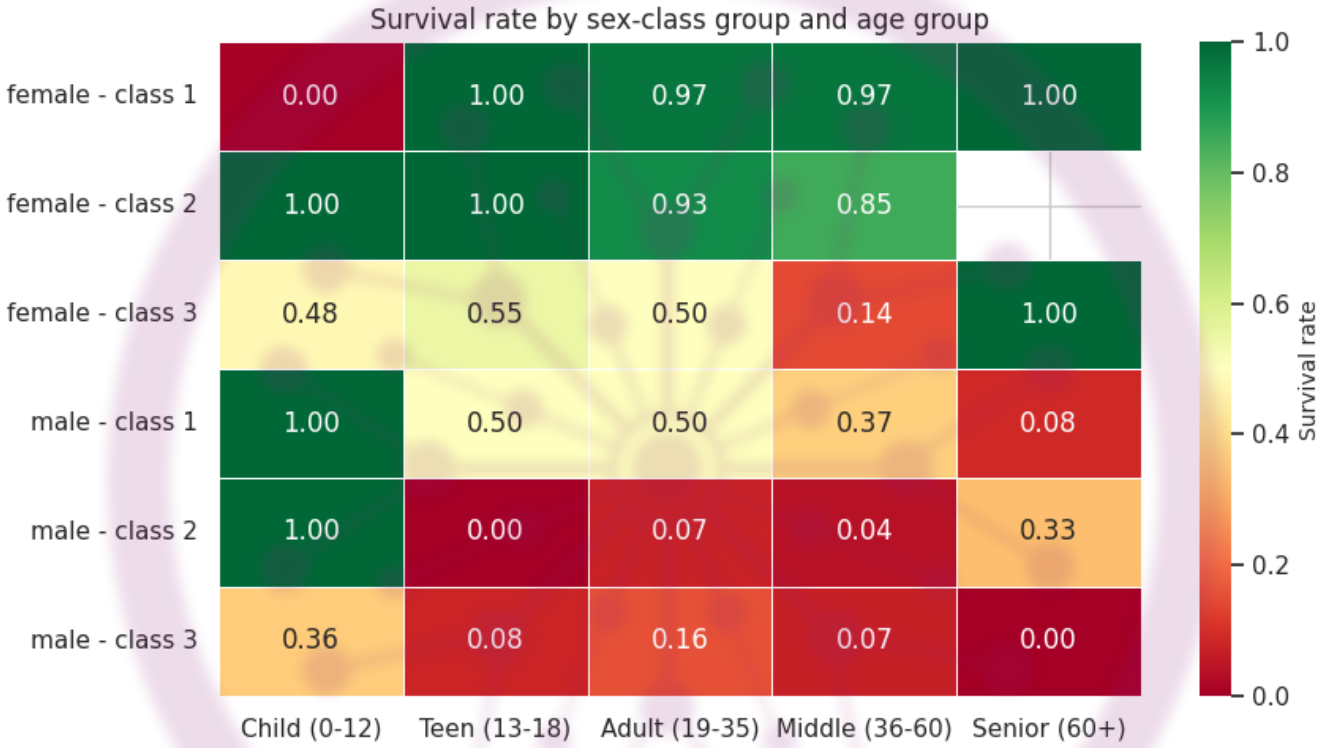
- **Caution:** Pearson correlation only captures linear relationships and the target is binary. `sex` is not in this heatmap (it is categorical) even though it is the strongest predictor, and `age`/`family_size` have non-linear effects that low r values hide. `pclass` and `fare` carry overlapping information (multicollinearity).

6. Multivariate EDA

6.1 Survival rate across class x sex x age group

Code

```
piv = d.pivot_table(index="sex_pclass", columns="age_group", values="survived", aggfunc="mean")
cnt = d.pivot_table(index="sex_pclass", columns="age_group", values="survived", aggfunc="count")
plt.figure(figsize=(9, 5))
sns.heatmap(piv, annot=True, fmt=".2f", cmap="RdYlGn", vmin=0, vmax=1, linewidths=.5, cbar_kws={"label":
 "Survival rate"})
plt.title("Survival rate by sex-class group and age group"); plt.ylabel(""); plt.xlabel("")
plt.tight_layout(); plt.show()
print("Number of passengers in each cell:")
print(cnt.fillna(0).astype(int))
```



Output

```
Number of passengers in each cell:
age_group         Child (0-12)  Teen (13-18)  Adult (19-35)  Middle (36-60)  \
sex_pclass
female - class 1             1            10             36              36
female - class 2             8             6             40              20
female - class 3            23            20             44              14
male - class 1               3             2             30              54
male - class 2               9             6             54              27
male - class 3              25            26            154              44

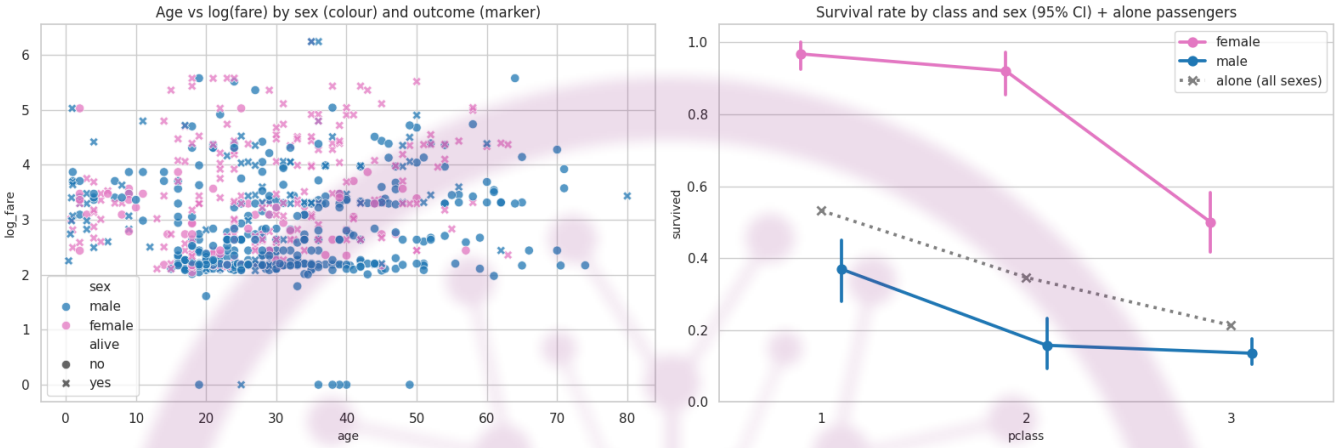
age_group         Senior (60+)
sex_pclass
female - class 1             2
female - class 2             0
female - class 3             1
male - class 1              12
male - class 2               3
male - class 3               4
```

Interpretation. The heatmap shows that combining variables reveals sharper groups than any single variable: women in First and Second class survive at nearly every age (about 0.9 to 1.0), while men in Second and Third class survive at very low rates after childhood. Boys in the youngest group do much better than adult men of the same class, which supports the "women and children first" policy. Some cells have only a handful of passengers (see counts), so those percentages are noisy.

6.2 Age, fare, sex and outcome together

Code

```
fig, ax = plt.subplots(1, 2, figsize=(16, 5.5))
sns.scatterplot(data=d, x="age", y="log_fare", hue="sex", style="alive",
                palette={"male": "#1f77b4", "female": "#e377c2"}, s=55, alpha=0.75, ax=ax[0])
ax[0].set_title("Age vs log(fare) by sex (colour) and outcome (marker)")
sns.pointplot(data=d, x="pclass", y="survived", hue="sex", ax=ax[1],
              palette={"male": "#1f77b4", "female": "#e377c2"}, errorbar=("ci", 95), dodge=0.2)
sns.pointplot(data=d[d.alone], x="pclass", y="survived", ax=ax[1], color="gray", linestyles=":",
              markers="x", errorbar=None, label="alone (all sexes)")
ax[1].set_ylim(0, 1.05); ax[1].set_title("Survival rate by class and sex (95% CI) + alone passengers")
ax[1].legend()
plt.tight_layout(); plt.show()
```

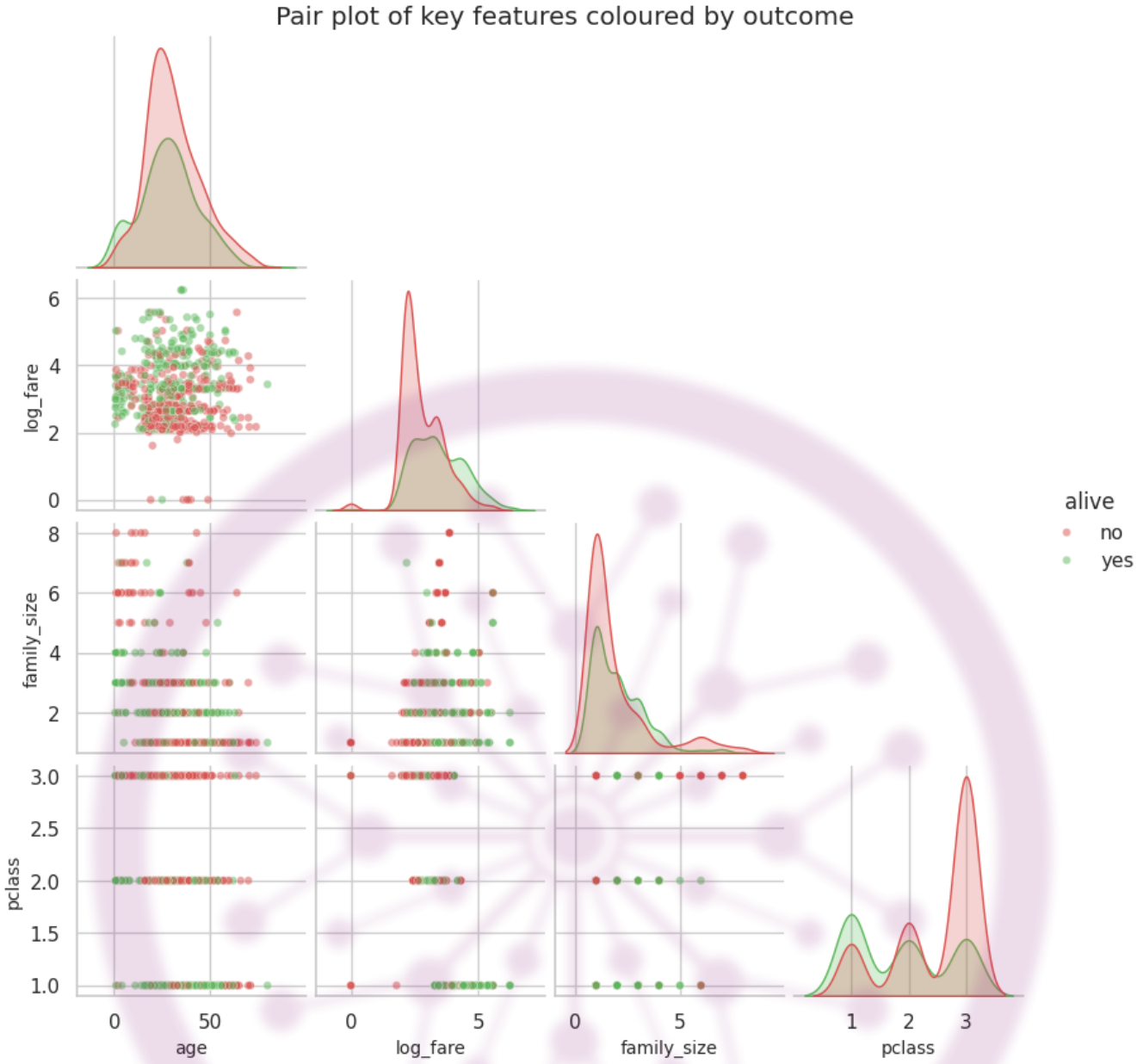


Interpretation. Female passengers (pink) are mostly marked as survivors across the age-fare plane, whereas blue male points are mostly non-survivors except at young ages or very high fares. The point plot makes the class-sex interaction clear: the gap between women and men is huge in First and Second class, and shrinks in Third class because third-class women fared much worse. Confidence intervals are widest for First-class men and women because of smaller group sizes.

6.3 Pair plot of key numerical features

Code

```
sub = d[["age", "log_fare", "family_size", "pclass", "alive"]].dropna()
g = sns.pairplot(sub, hue="alive", hue_order=["no", "yes"], palette=["#d9534f", "#5cb85c"],
                 diag_kind="kde", plot_kws={"alpha": 0.5, "s": 22}, height=2.1, corner=True)
g.fig.suptitle("Pair plot of key features coloured by outcome", y=1.01)
plt.show()
```



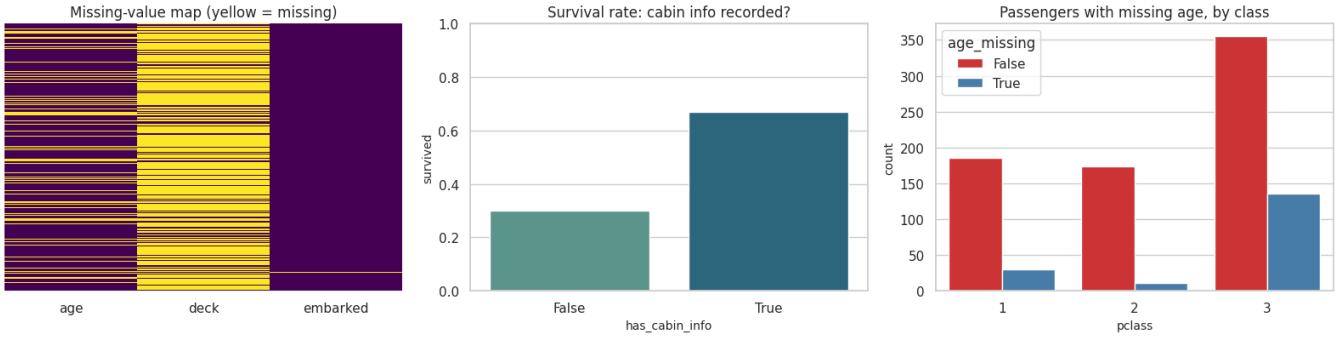
Interpretation. In the diagonal densities, survivors are shifted toward higher `log_fare` and lower `pclass` (better class), while age densities largely overlap apart from the child peak. The `age` vs `pclass` panel shows that First class passengers are older on average. No single pair separates the two outcome colours cleanly, which confirms that survival depends on **combinations** of features, not one variable.

7. Patterns, Trends, and Anomalies

7.1 Missing values as a pattern

Code

```
fig, ax = plt.subplots(1, 3, figsize=(16, 4.2))
sns.heatmap(d[["age", "deck", "embarked"]].isna(), cbar=False, yticklabels=False, cmap="viridis",
ax=ax[0])
ax[0].set_title("Missing-value map (yellow = missing)")
sns.barplot(data=d, x="has_cabin_info", y="survived", ax=ax[1], palette="crest", errorbar=None)
ax[1].set_title("Survival rate: cabin info recorded?"); ax[1].set_ylim(0, 1)
sns.countplot(data=d, x="pclass", hue="age_missing", ax=ax[2], palette="Set1")
ax[2].set_title("Passengers with missing age, by class")
plt.tight_layout(); plt.show()
print(d.groupby("has_cabin_info")["survived"].agg(["mean", "count"]).round(3))
print(d.groupby("age_missing")["survived"].agg(["mean", "count"]).round(3))
print("Share of missing-age passengers by class:",
d[d.age_missing]["pclass"].value_counts(normalize=True).round(3).to_dict())
```



Output

```
                mean  count
has_cabin_info
False           0.299    688
True            0.670    203
              mean  count
age_missing
False        0.406    714
True         0.294    177
Share of missing-age passengers by class: {3: 0.768, 1: 0.169, 2: 0.062}
```

Interpretation. Missingness is **not random**. Passengers with a recorded cabin survived at 67.0% against 29.9% for those without, because cabin records exist mostly for First class (and for survivors who could be interviewed later). 76.8% of the passengers with missing age are in Third class, and missing-age passengers survived less (29.4% vs 40.6%). So simple mean-imputation can hide information; a "missing" flag and class-wise median imputation is a better modelling choice.

7.2 Outliers and unusual observations

Code

```
q1, q3 = d["fare"].quantile([.25, .75]); fence = q3 + 1.5*(q3 - q1)
print("Fare IQR upper fence: %.2f | passengers above fence: %d" % (fence, (d.fare > fence).sum()))
print("\nTop 6 fares:")
print(d.nlargest(6, "fare")[["fare", "pclass", "sex", "age", "survived", "embark_town"]].to_string())
print("\nPassengers with fare = 0: %d (survival rate %.2f)" % ((d.fare == 0).sum(), d[d.fare ==
0].survived.mean()))
print(d[d.fare == 0][["pclass", "sex", "age", "survived"]].pclass.value_counts().to_dict())
print("\nOldest passengers (age >= 65):")
print(d[d.age >= 65][["age", "sex", "pclass", "survived"]].sort_values("age").to_string())
print("\nInfants (age < 1):")
print(d[d.age < 1][["age", "sex", "pclass", "survived"]].to_string())
print("\nLargest families:")
print(d.nlargest(5, "family_size")[["family_size", "pclass", "sex", "age", "survived"]].to_string())

fig, ax = plt.subplots(1, 2, figsize=(14, 4.5))
sns.stripplot(data=d, x="pclass", y="fare", hue="alive", hue_order=["no", "yes"], palette=["#d9534f",
"#5cb85c"],
              alpha=.6, jitter=.25, ax=ax[0])
ax[0].axhline(fence, color="k", ls="--", lw=1, label="IQR fence"); ax[0].set_title("Fare by class -
outliers highlighted"); ax[0].legend()
sns.lineplot(data=d.assign(age_bin=(d.age // 5 * 5)).dropna(subset=["age_bin"]), x="age_bin",
y="survived",
             hue="sex", marker="o", palette={"male": "#1f77b4", "female": "#e377c2"}, errorbar=None,
ax=ax[1])
ax[1].set_title("Trend: survival rate across age (5-year bins)"); ax[1].set_xlabel("age (bin start)")
plt.tight_layout(); plt.show()
```

Output

```
Fare IQR upper fence: 65.63 | passengers above fence: 116

Top 6 fares:
         fare  pclass     sex   age  survived  embark_town
258  512.3292       1  female  35.0         1    Cherbourg
679  512.3292       1    male  36.0         1    Cherbourg
737  512.3292       1    male  35.0         1    Cherbourg
27   263.0000       1    male  19.0         0  Southampton
88   263.0000       1  female  23.0         1  Southampton
341  263.0000       1  female  24.0         1  Southampton

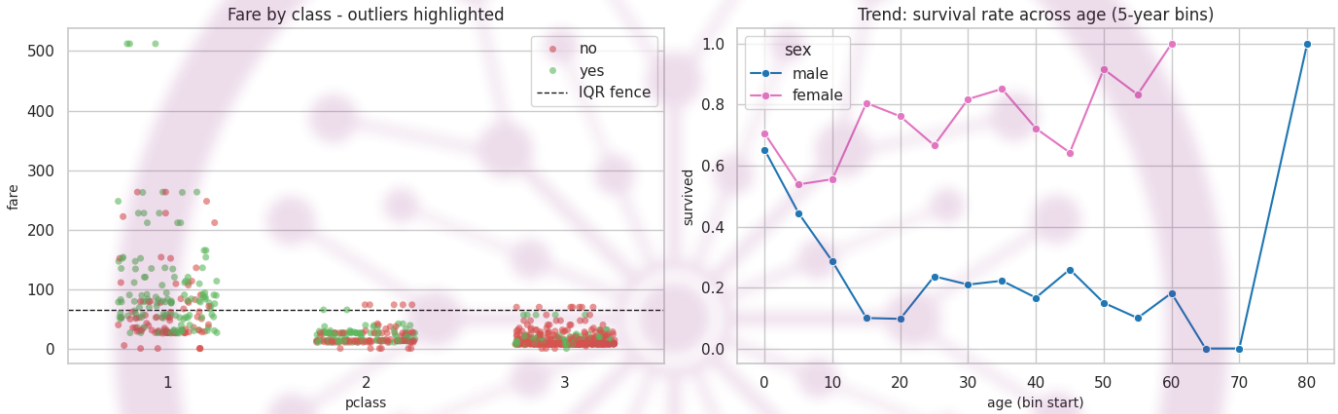
Passengers with fare = 0: 15 (survival rate 0.07)
{2: 6, 1: 5, 3: 4}

Oldest passengers (age >= 65):
      age   sex  pclass  survived
54   65.0  male       1         0
```

```
280  65.0  male       3         0
456  65.0  male       1         0
33   66.0  male       2         0
745  70.0  male       1         0
672  70.0  male       2         0
116  70.5  male       3         0
96   71.0  male       1         0
493  71.0  male       1         0
851  74.0  male       3         0
630  80.0  male       1         1

Infants (age < 1):
      age     sex  pclass  survived
78   0.83    male       2         1
305  0.92    male       1         1
469  0.75  female       3         1
644  0.75  female       3         1
755  0.67    male       2         1
803  0.42    male       3         1
831  0.83    male       2         1

Largest families:
     family_size  pclass     sex  age  survived
159           11       3    male  NaN         0
180           11       3  female  NaN         0
201           11       3    male  NaN         0
324           11       3    male  NaN         0
792           11       3  female  NaN         0
```



Findings: trends, anomalies and interesting patterns.

- **Fare outliers:** 116 passengers pay more than the 65.6 fence. Three tickets cost exactly 512.33 (the maximum), about 8x the First-class median - extreme, but all three are First class and all survived, so they are real, high-value observations. Keep them but use log-fare or robust scaling.
- **Zero fares:** 15 passengers (all men, mostly First and Second class) have fare = 0 and only about 6.7% survived. These are likely crew/guests or missing price records - an unusual group worth flagging.
- **Age trend:** for women, survival stays high across most ages; for men, survival is highest for boys and drops sharply for adults. All 7 infants under 1 year survived, while 10 of 11 passengers aged 65+ died (the only exception is the 80-year-old).
- **Family size anomaly:** families of 8 and 11 (13 passengers in total) had 0% survival, the opposite of small families.
- **Hidden structure:** the strong sex x class interaction (women in First/Second class ~92-97% survival vs men in Second/Third class ~14-16%).

8. Final Findings

1. **Sex is the most powerful predictor.** 74.2% of women survived versus 18.9% of men - the largest gap among all features (consistent with a "women and children first" evacuation).
2. **Passenger class is the second key driver.** Survival falls from 63.0% (First) to 47.3% (Second) to 24.2% (Third); fare is highly correlated with class (r = -0.55), so they carry overlapping information.
3. **Sex and class interact.** First/Second-class women survived at 92-97%, third-class women at 50%, and men in Second/Third class at only 13-16%. Tree-based models or explicit interaction terms should capture this.

4. **Age matters mainly at the extremes.** Children 0-12 survived at 58% and every infant under 1 survived, whereas passengers over 60 survived at 22.7%; overall correlation of age with survival is only -0.08 because the effect is non-linear.
5. **Family size has a non-linear effect.** Small families (2-4 people) survived far more than solo travellers (30.4%), but large families (5+) did poorly - bin or use categorical family-size features.
6. **The target is mildly imbalanced (61.6% vs 38.4%).** Use stratified splits and evaluate with F1, recall and ROC-AUC rather than accuracy alone.
7. **Fare is heavily right-skewed with real outliers** (skewness 4.79, 116 values above the IQR fence). A log transform brings skewness down to 0.39; do not delete the outliers because they are valid first-class tickets.
8. **Missing data is informative.** 77.2% of `deck` and 19.9% of `age` are missing, mostly for Third-class passengers. Impute age by class/sex median, add missing-indicator flags, and avoid using `deck` directly (or use only a "cabin known" flag).

How this helps in modelling: drop redundant columns (`alive`, `class`, `embarked`, `who`/`adult_male` after feature engineering to avoid leakage-like duplication), encode `sex`/`embark_town`, create `family_size`, `is_alone`, `age_group`, `log_fare` and missing flags, then compare logistic regression (baseline) with Random Forest / Gradient Boosting that can learn the interactions found here.

Submission Note

The accompanying notebook (Titanic_EDA.ipynb) contains the same code, outputs and graphs as this PDF. Upload it (and optionally this PDF) to a public GitHub repository or Zenodo record and submit the public link on LMS.

Dataset source: Kaggle Titanic - Machine Learning from Disaster (https://www.kaggle.com/c/titanic); copy used: https://github.com/mwaskom/seaborn-data